01 // PRESENTATION

Commercializing Zero-Knowledge: Realities from the Field

Deconstructing enterprise & GovTech hypotheses to discover practical heuristics

Zero-Knowledge in the Real World: Lessons from the Trenches

An open post-mortem exploring why enterprise ZK pitches hit unexpected walls, what database fragmentation taught us, and how to rethink ZK product-market fit from first principles.

SPEAKER BACKGROUND

Ex-Senior Product Owner @ 1Matrix
Led product discovery for blockchain infrastructure & enterprise pilots.

Vietnam Blockchain Service Network (VBSN)
Pushed national-level infrastructure and cross-company attestation layers.

Government Cipher Committee Engagement
Lobbied & evaluated ZK proving scheme feasibility with *Ban Cơ yếu Chính phủ*.

Pushing ZK Technology in Vietnam

The theoretical promise vs. the primary hurdles encountered in the field

THE PROMISE: PROVING WITHOUT REVEALING

The Core Capability

The Mechanism: ZK technology fundamentally allows a party to prove that a statement is valid without revealing the underlying data.

The Real-World Need: In Vietnam, data laws prevent companies from sharing full information of clients, while citizens must reveal certificates to get notarized copies.

The Initial Fit: ZK seemed like a natural fit to eliminate repeated verification and protect personal information without transferring raw data.

THE REALITY: TWO MAJOR HURDLES

What the Field Revealed

1. Database Fragmentation (The Biggest Hurdle): Whether in corporate groups or local government entities, underlying databases were so fragmented that unifying them or establishing a baseline was cost-prohibitive.

2. Government Sector Approval: ZK is a new proving scheme requiring national approval from the Government Cipher Committee, demanding multiple rounds of effect studies before any real deployment.

Overview of this sharing: Unpacking the 2 specific use cases we explored, the regulatory barrier, and questions to explore new heuristics.

Case 1: Unified Verification in a Mother Group

Can verification data at one company be used for all others without revealing data?

INITIAL EXPECTATION

"ZK Seems Like a Good Fit"

The Legal Barrier: Different companies in the same mother group in Vietnam cannot share full information of a client to one another due to data law.

The User Friction: Users have to perform re-verification every time they interact with a different company in the group.

The Core Idea: Can verification data at one company be used for all others without revealing data? ZK can prove without revealing, seeming like the ideal solution.

REALITY DISCOVERED

Database Fragmentation Was the Hurdle

Structural Mismatch: Even within the same mother group, different companies had completely different database structures.

The Burden: Cleaning and unifying all the different database structures required far too much effort.

The Commercial Verdict: The cost to clean and unify database structures is not justified just for saving user time.

Takeaway: Cryptography couldn't compensate for the immense cost of unifying fragmented company databases.

04 // GOVTECH CASE STUDY

Case 2: ZK for Public Notary

Proving certificate validity without revealing personal information

INITIAL EXPECTATION

Citizen Registry for All Certificates

The Status Quo: A person has to reveal certificate information to get notarized copies by the authority.

The ZK Vision: If that person has a registry for all certificates, they can prove to companies without revealing personal information.

The Key Prerequisite: This is only viable if there is a government entity with all the citizen certificate data first to enable a ZK layer on top.

REALITY DISCOVERED

Local Government Databases Were Too Fragmented

The Investigation: We explored where citizen certificate data actually resided across local government entities.

The Ground Reality: Databases in local government entities were so fragmented, that no one has all of it.

The Structural Barrier: Without a government entity holding complete citizen certificate data, there was no foundational layer to build a ZK layer on top of.

Takeaway: You need an authoritative government database first before an attestation layer can exist.

The Institutional Wall: Innovation vs. Cryptographic Sovereignty

Navigating the Government Cipher Committee (Ban Cơ yếu Chính phủ)

Comparison Criteria	Web3 / Commercial Startups	Government Cipher Committee (Ban Cơ yếu)
Primary Objective	Speed to market, rapid developer iteration, user adoption, frictionless product UX.	**National Sovereignty:** Uncompromised resilience against foreign cryptanalysis, quantum threats, and state adversaries.
Attitude Toward Novel Math	Eager to adopt bleeding-edge proving schemes (PLONK, Halo2, STARKs) and novel elliptic curves (BN254, BLS12-381).	**Extreme Mathematical Conservatism:** Novel curves require exhaustive multi-round effect studies, formal vulnerability analysis, and committee reviews.
Timeline Cycles	**Weeks to Months:** Fast sprints, hackathons, and quarterly product milestones.	**Multi-Year Evaluations:** Formal standards defense, national security vetting, and legislative decree updates.
Approval Prerequisite	"Does the circuit compile and pass automated test suites?"	"Has this proving scheme been formally certified for sovereign infrastructure usage?"

Key Discovery: In sovereign and regulated sectors, cryptographic conservatism always trumps technological novelty. Startups run out of runway while sovereign accreditation is just starting.

Discussion 1: Applying Functional Analysis to ZK

Deconstructing the technology: What atomic functions did we completely miss?

Methodology: Functional Analysis strips away marketing buzzwords to isolate what a system fundamentally *does* at an atomic level.

THE FUNCTION WE BET ON

Input Concealment

"Proving Without Revealing" (Zero-Knowledge Masking)

Proving that a private witness w satisfies predicate $P(x, w) = true$ without disclosing any data about w to the verifier.

Our Pilot Reality: We bet 100% of our enterprise roadmap on this single function. But Input Concealment strictly requires standardized, authoritative data inputs—and collided directly with fragmented legacy databases.

OPEN QUESTION TO THE ROOM

What are the OTHER basic functions of ZK?

(Even if they seem obvious or well-known to you!)

When you decompose Zero-Knowledge systems down to their irreducible mathematical primitives, what other elementary capabilities exist?

Inquiry for Practitioners: When you strip away marketing, what other fundamental functions does ZK perform that our enterprise pilots completely overlooked?

Opening the floor: What other basic functions or capabilities does ZK have?

Opening the Floor: Questions for Builders & Practitioners

Applying the Perspective-Heuristics method to explore where practical ZK adoption lives

Perspective-Heuristics Method: Breakthroughs happen by shifting our *Perspective* (how we represent the problem space) and refining our *Heuristics* (rules of thumb for evaluating fit).

QUESTION 01 // HEURISTICS FOR FIT

What mental filters or heuristics do you use to evaluate whether a problem genuinely warrants ZK?

When someone pitches a new ZK use case, how do you determine if it is a genuine cryptographic necessity or expensive over-engineering? Where is the boundary between an API/database solution and a true ZK problem?

Points to Consider:

- Does ZK belong strictly at adversarial boundaries (zero mutual trust)?
- Can internal enterprise compliance ever justify proof generation overhead?
- How do you assess data readiness before committing to circuit design?

QUESTION 02 // FIELD PROBLEMS & PERSPECTIVES

What real-world field problems can be cleanly solved with ZK's basic functions?

Beyond the classic tropes of identity (DID) and notarizing diplomas, what bottlenecks in your industry possess naturally clean, digital inputs where verification is expensive or trust is scarce?

Domains to Explore:

- Verifiable AI: proving model weights or inference execution without disclosing data
- Financial compliance: proving solvency or risk limits without sharing trade books
- Supply chain & telemetry: hardware-attested sensor data and provenance